Securing the Future: A Literature Review on Post-Quantum and Quantum Cryptography

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December 15th, 2025

Abstract

The possible rise of quantum computers poses an existential threat to the global digital infrastructure, rendering foundational public-key cryptosystems insecure against Shor's algorithm. With recent industry analysis indicating that these vulnerable standards currently dominate the field of cryptography, the risk of "Harvest Now, Decrypt Later" (HNDL) attacks necessitates a strategic turn. This literature review critically evaluates the two primary paradigms emerging to address this challenge: Post-Quantum Cryptography (PQC) and Quantum Key Distribution (QKD). We analyze PQC protocols such as Crystals-Kyber, which offers computational security compatible with existing classical infrastructure. In contrast, we examine QKD protocols such as BB84, which derive information-theoretic security from the fundamental laws of quantum mechanics rather than mathematical assumptions. Through a comparative analysis of urgency, security foundations, and deployment feasibility, this review concludes that while QKD offers superior long-term security guarantees for high-value critical infrastructure, PQC represents the only scalable and secure solution that can be used as an immediate measure.

Keywords: Post-Quantum Cryptography, Quantum Key Distribution, Harvest Now, Decrypt Later, Computational Hardness Assumptions, Information-Theoretic Security, Hybrid Cryptographic Architecture

1 Introduction

In today's digital landscape, society relies heavily on public-key encryption to secure sensitive information. Important infrastructure like banking systems, email communications, digital transactions, and web server authentication depends on cryptographic standards such as RSA [1] and Elliptic Curve Cryptography (ECC) [2] to ensure data confidentiality and authenticity. Recent industry analysis highlights the scale of this dependency; a 2025 survey by SandboxAQ reveals that RSA still underpins approximately 11.5 million public keys on the internet, while ECC accounts for roughly 6 million [3].

However, these established standards face an existential threat from the advent of quantum computing. The RSA algorithm, which has served as a cornerstone of digital security for decades, relies on the computational complexity of integer factorization, the difficulty of factoring the product of two large prime numbers. While current classical computers lack the computational power to reverse this process in a reasonable timeframe, this assumption no

longer holds in the quantum realm. As detailed in Peter Shor's seminal 1994 paper[4], Shor's Algorithm demonstrates that a quantum computer, that uses the principles of superposition and entanglement, could solve the integer factorization problem in polynomial time. The availability of a sufficiently powerful quantum computer would destroy the current public-key infrastructure, allowing adversaries to decrypt sensitive communications and forge digital signatures with ease.

In response to these emerging threats, the scientific community is actively pursuing two distinct cryptographic approaches to preserve digital security: Post-Quantum Cryptography (PQC) and Quantum Cryptography.

Post-Quantum Cryptography refers to new cryptographic algorithms that run on classical hardware but rely on complex mathematical problems such as structured lattices that are believed to be intractable even for quantum computers. While numerous PQC candidates have been proposed, this review focuses on algorithms that have withstood rigorous cryptanalysis and are currently being standardized by the National Institute of Standards and Technology (NIST).[5]

In contrast, Quantum Cryptography, primarily realized through Quantum Key Distribution (QKD), does not rely on mathematical assumptions entirely. Instead, it derives its security guarantees from the fundamental laws of quantum mechanics, ensuring that any attempt to intercept the key introduces detectable errors.

Despite the fact that quantum computers have not yet been properly developed, the urgency to transition to quantum-secure paradigms is driven by the 'Harvest Now, Decrypt Later' (HNDL) threat [6]. With this attacks adversaries can intercept and store encrypted traffic today with the intention of decrypting it once quantum capabilities become available. Consequently, the confidentiality of long-term secrets is already compromised by the future existence of quantum hardware. This retrospective threat shows the need of evaluation and deployment of quantum-resistant schemes.

The remainder of this review is organized as follows: Section 2 details the possible cryptographic solutions for the quantum era, divided into two subsections. Subsection 2.1 examines Post-Quantum Cryptography, with a specific focus on the lattice-based CRYSTALS-Kyber algorithm standardized by NIST. Subsection 2.2 explores Quantum Key Distribution (QKD), detailing its reliance on physical laws to secure key exchange. Section 3 presents a comparative analysis of these two paradigms, evaluating them across three critical dimensions: the urgency of adoption and standardization defined by the "Mosca Inequality", the fundamental distinction between computational complexity and information-theoretic security guarantees, and the practical deployment challenges regarding infrastructure and system adaptability. Finally, Section 4 summarizes the findings and provides a final verdict on the future of cryptographic security.

2 Cryptographic Schemes

This section provides a technical overview of the two primary cryptographic paradigms emerging to address the quantum threat. First, we examine Post-Quantum Cryptography (PQC),

which focuses on developing new mathematical algorithms secure against quantum cryptanalysis. Second, we explore Quantum Key Distribution (QKD), which utilizes the physical properties of quantum mechanics to secure communications.

2.1 Post-Quantum Cryptography (PQC)

Post-Quantum Cryptography (PQC) refers to cryptographic algorithms designed to run on widely available classical hardware while remaining secure against cryptanalysis by both classical and quantum computers. These algorithms rely on computational hardness assumptions and complex mathematical problems that are believed to be intractable even for adversaries armed with powerful quantum algorithms like Shor's algorithm or Grover's algorithm. [7]

While current public-key standards such as RSA and ECC depend on the difficulty of integer factorization and discrete logarithms, PQC algorithms utilize mathematical structures that have no known exponential speedup in the quantum domain. The field is generally categorized into four primary families: Lattice-based cryptography, Code-based cryptography, Multivariate cryptography, and Hash-based cryptography. [8] Among these, lattice-based cryptography has been prioritized by standardization bodies due to its strong security guarantees and efficiency.

2.1.1 Lattice-Based Cryptography and CRYSTALS-Kyber

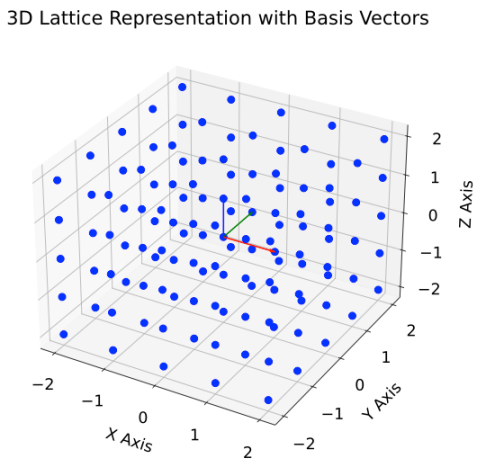

Figure 1: **3D Lattice Representation.** A visualization of a lattice structure in three-dimensional space with basis vectors. In the high-dimensional spaces used by Kyber (e.g., 512 dimensions), finding the nearest lattice point to a given random location in this space becomes exponentially difficult. (Source: KyberLib [9])

Lattice-based cryptography relies on the geometric structure of lattices. The security of these systems typically stems from the difficulty of the "Shortest Vector Problem" (SVP) or the "Closest Vector Problem" (CVP).[10] While trivial in low dimensions as shown in Figure 1, in high dimensions, finding the nearest lattice point to a given random point is computationally infeasible, providing the hardness assumption that underpins the scheme's security.

The primary standard selected by NIST for general-purpose public-key encryption is CRYSTALS-Kyber. [11] Kyber operates on a specialized variant of the lattice problem known as Module Learning With Errors (M-LWE). (see Figure 2) In this scheme, the underlying math involves solving a system of linear equations over a ring structure where small, secret "errors" (noise) are deliberately introduced. An adversary's goal is to distinguish these noisy samples from truly random data—a task believed to be intractable for quantum computers.

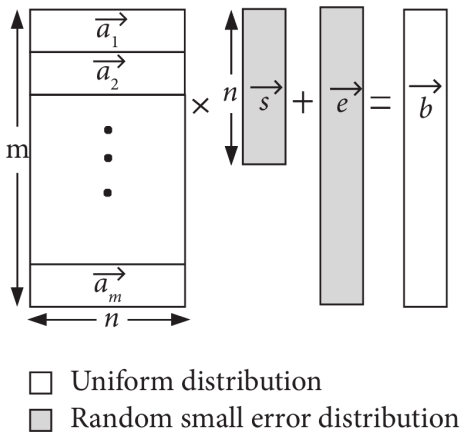

Figure 2: **Learning With Errors (LWE) Mechanism.** A visual representation of the mathematical problem underlying Kyber. The public vector components ($\vec{a}$) are multiplied by the secret vector ($\vec{s}$), and a small error vector ($\vec{e}$) is added to produce the public output ($\vec{b}$). Here, $\vec{s}$ serves as the **private key**, which is generated prior to this operation. The error term $\vec{e}$ effectively "masks" $\vec{s}$, making it computationally intractable to recover the private key (which is the private vector $\vec{s}$) from the public output $\vec{b}$. (Source: Cheng et al. [12])

To achieve high performance, Kyber utilizes the Number Theoretic Transform (NTT) [13] for polynomial multiplication. This allows the algorithm to perform key generation and encapsulation in microseconds, making it highly efficient for real-world applications ranging from web servers to smart cards. To accommodate varying security requirements, the algorithm is defined with three distinct versions(Kyber-512, Kyber-768, and Kyber-1024). Each of them offer differing level of protection and these correspond roughly to the security strengths of AES-128, AES-192, and AES-256, effectively benchmarking them against the Advanced Encryption Standard (AES),[14] which serves as the current global standard for secure symmetric encryption.

2.2 Quantum Key Distribution (QKD)

Quantum Cryptography, primarily realized as Quantum Key Distribution (QKD), is a paradigm of secure communication that diverges fundamentally from classical methods. Unlike Post-Quantum Cryptography, which relies on computational hardness assumptions, QKD derives its security guarantees from the fundamental laws of quantum physics. Its most noticeable characteristic is the ability to provide information theoretic security, meaning that the generated keys are provably secure against an adversary with unlimited computing power, provided the laws of quantum mechanics hold true.

The security mechanism of QKD operates using two core principles: the Heisenberg Uncertainty Principle [15] and the No-Cloning Theorem. [16] These laws dictate that it is physically impossible to create an exact copy of an unknown quantum state, and that any attempt to measure a quantum system inevitably disturbs it. In practice, information is encoded into the quantum states of particles, such as the polarization of single photons. If an eavesdropper attempts to intercept or measure these photons during transmission, the state is altered. This disturbance introduces detectable errors into the key exchange, alerting the legitimate parties to the breach and allowing them to discard the compromised data.

2.2.1 Protocol Example: BB84

The most widely established QKD protocol is BB84 (see Figure 3), proposed by Bennett and Brassard in 1984. In this scheme, Alice transmits bits encoded in non-orthogonal quantum bases (e.g., rectilinear and diagonal polarizations). Bob measures these incoming photons using a random choice of bases.

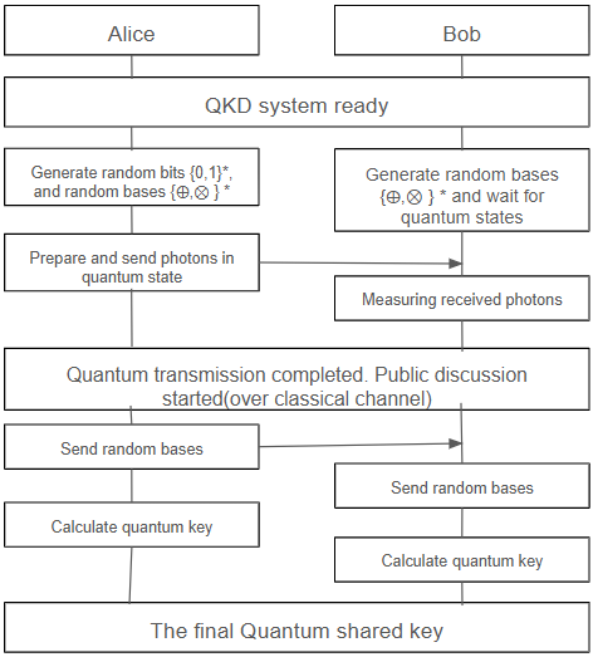

Figure 3: **BB84 Protocol Mechanics.** Alice sends qubits through a quantum channel. Eve's attempt to measure the photons collapses the quantum state, introducing errors that Bob detects during the sifting process. (Source: Recreated Image from Abdullah and Obeid [17])

After the transmission, Alice and Bob communicate over an authenticated public channel to compare which bases they used, its process is known as "sifting." [18] They retain only the bits where their bases matched. Since an eavesdropper cannot know the correct basis in advance, any interception attempt effectively guesses the basis, resulting in a statistically significant error rate that reveals the presence of the eavesdropper.

2.2.2 Advanced Security and Infrastructure

To reduce reliance on trusted hardware, recent research has focused on Device-Independent protocols. [19] These methods utilize the violation of Bell Inequalities(see Figure 4) to mathematically certify that the generated keys are private and random, even if the internal working of the device is unknown or untrusted.

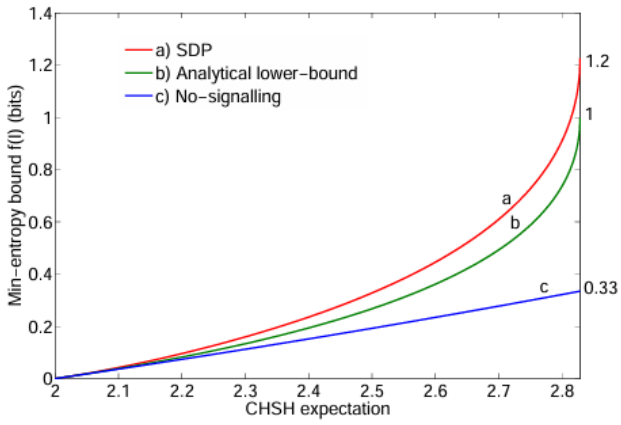

Figure 4: **Bell Inequality Violation and Certified Randomness.** A plot of the min-entropy (certified randomness) against the CHSH expectation (Bell violation). The graph demonstrates that as the observed violation increases beyond the classical limit of 2, the amount of trusted, private randomness guaranteed by the laws of physics increases. (Source: Pironio et al. [20])

Deployment of QKD requires specific physical infrastructure, primarily a quantum channel (such as optical fiber or free-space optical links) to transmit the fragile quantum states. Current technology is limited by signal loss over long distances. In addition, active components that connects the optical fibres do not support quantum information. To overcome this, future global-scale networks will require the development of Quantum Repeaters which are devices capable of distributing entanglement over long distances without measuring the quantum information. [21]

3 Comparative Analysis

The choice between Post-Quantum Cryptography (PQC) and Quantum Key Distribution (QKD) is not merely a technical decision but a strategic one, governed by the constraints of time, security requirements, and infrastructure costs. This section compares the two groups of schemes across three critical dimensions: the urgency of adoption, the fundamental nature of their security, and the practical challenges of deployment.

3.1 Urgency, Maturity, and Standardization

The most pressing factor driving the adoption of quantum-secure schemes is the timeline of the threat. This urgency is often conceptualized by the "Mosca Inequality": $D + T > Q_c$, where

D is the duration for which data must remain secure (shelf-life), T is the time required to migrate infrastructure, and Q_c is the time until quantum computers or some other method, breaks the currently deployed cryptographic schemes. [22] If this inequality holds, organizations are already late, and data encrypted today is vulnerable to HNDL attacks, where adversaries intercept traffic now to decrypt it once quantum hardware becomes available. In this context, PQC offers a significant advantage regarding the "Migration Time" (T). Because PQC algorithms like CRYSTALS-Kyber have been recently standardised by NIST and are compatible with existing protocols, they allow for a faster transition than QKD. QKD still falls behind when it comes to being standardised and requires physical hardware upgrades, which drastically increases the migration time. While it is true that establishing dedicated quantum channels might be feasible for specific high-value connections (e.g., between government agencies), scaling the standard for the entire global Internet infrastructure to support QKD is practically infeasible. Therefore, for immediate mitigation of the quantum threat, PQC is currently the only viable solution for widespread standardization.

3.2 Security Foundations: Computational Complexity vs. True Randomness

The fundamental distinction between the two paradigms lies in their source of trust. PQC relies on computational security; it assumes that specific mathematical problems (like Module-LWE in Kyber) are hard to solve. While these problems are believed to be intractable, they are not proven to be impossible to solve. A future mathematical breakthrough could theoretically break PQC algorithms without requiring a quantum computer, much like how improvements in factoring weakened RSA over time. For instance, the Rainbow signature scheme which was once a leading NIST finalist, was recently proven insecure against classical cryptanalysis. Beullens demonstrated that its underlying multivariate quadratic problem could be solved efficiently on a standard laptop,[23] highlighting the risk that hidden mathematical vulnerabilities may exist within PQC candidates regardless of quantum computing capabilities.

In contrast, QKD offers Information-Theoretic Security, which relies on the laws of physics rather than assumptions about computational power. QKD systems can generate keys with "certified randomness," where the security is guaranteed by the violation of Bell inequalities. [24] This ensures that the key exchange is perfectly secure against any adversary, regardless of their computing power. However, Renner and Wolf note a critical nuance: while the protocol of QKD is perfect, the physical implementation can suffer from side-channel vulnerabilities, whereas PQC implementations are generally more mature and better understood.

3.3 Deployment and Infrastructure Feasibility

The third area of comparison is the practicality of deployment. PQC utilizes a "software-only" approach, allowing it to be deployed over existing communication channels (fiber, Wi-Fi, 5G) and on common hardwares such as IoT devices. This supports cryptographic adaptability, enabling systems to update algorithms via software patches if a vulnerability is discovered.

QKD, however, faces significant infrastructure obstacles. It requires a dedicated Quantum Channel to transmit fragile quantum states. Because quantum signals cannot be amplified by

standard repeaters, current QKD links are limited by distance and signal loss. [25] Achieving a global QKD network would require ”trusted nodes” or the future development of Quantum Repeaters. Consequently, PQC is viewed as the scalable solution for the general internet, while QKD is currently best suited for high-security, which are needed in critical infrastructure.

4 Conclusion

The possible arrival of quantum computing requires a paradigm shift from the cryptographic approaches that have protected global communications for decades. As analyzed in this review, both PQC and QKD have their distinct features of computational adaptability and immutable physical guarantees, respectively. Since no single solution addresses every aspect of this threat, it is necessary to evaluate the distinct cases where each paradigm excels.

The choice between PQC and QKD depends on the urgency of the situation, the security foundation, and the constraints on infrastructure and scalability, as mentioned in Section 3 . This review focused on these cases in terms of short-term versus long-term timelines and situations of deployment.

In the long term, it is essential for QKD to be established as the core standard for key distribution. PQC protocols rely on computational hardness assumptions, which presents a long-term vulnerability; cryptanalytic breakthroughs can occur at any time, rendering the current assumptions of PQC protocols obsolete. Just as the Rainbow signature scheme was expected to be a NIST PQC standard but was broken by classical computers, it is not impossible for protocols like Kyber to be broken in the future. However, replacing the general internet or the IoT with QKD would face a prohibitive timeline. Hence, in the short term, PQC protocols should be the standard until a ubiquitous migration to QKD takes place.

However, it may also be the case that countries or worldwide organizations cannot fully migrate to QKD infrastructure due to challenges like cost. In such cases, it is important to consider QKD specifically for critical sectors. High-assurance environments where data must remain secret for decades (e.g. state secrets or bank secrets) are areas that require information-theoretic security. Only in such critical infrastructure can QKD be deployed to ensure key exchange, allowing data to be kept safe and secure. For IoT or the general internet, PQC will be a much more viable option, as QKD poses logistical barriers and significant deployment costs.

Ultimately, the future of digital security does not lie in a single approach, but in leveraging the strengths of both paradigms through a Hybrid Cryptographic Architecture. PQC protocols should serve as the agile and ubiquitous layer, securing general internet traffic across the global network. Simultaneously, QKD should be selectively deployed to secure the internet’s backbone, ensuring that the world’s most critical data channels remain secure against adversaries possessing quantum computing power. By integrating the computational scalability of mathematical algorithms with the certified randomness of quantum mechanics, this hybrid approach offers the most robust and resilient defense for the post-quantum era.

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